

WELCOME TO OUR PRESENTATION

Our Project

**“Machine Learning for Predictive
Analytics in Healthcare: Patient
Appointments and Physician
Billing”**

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Agenda

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Introduction



- ❖ Simplify the process of scheduling appointments between patients and healthcare providers.



- ❖ Two key machine learning components: Doctor's Fee Prediction System and No-Show Prediction Model.



- ❖ Doctor's Fee Prediction System analyzes factors like specialization, experience, location, and patient demand to recommend competitive fees.



- ❖ No-Show Prediction Model forecasts likelihood of missed appointments based on historical data and patient behavior.

Motivation



❖ Improved patient experience



❖ Increased efficiency



❖ Enhanced patient engagement



❖ Enhance convenience and fairness in the appointment process.



❖ Improve resource allocation and reduce no-shows.

Objectives



Create an efficient and user-friendly online platform that allows patients to easily book appointments with doctors.



Provide accurate fee recommendations for fairness and competitiveness.



Leverage historical data and patient behavior to forecast appointment no-shows.



Enable proactive measures to address potential no-shows and optimize schedules.

Related Work (Literature Review)

Yavuz et al. [3] The study evaluates the interface of the mobile Centralized Doctor Appointment System (CDAS), the largest mobile e-government health application in Turkey. The system facilitates citizens in making medical appointments at public hospitals. The evaluation aims to assess the usability and functionality of the CDAS interface, considering its significance in enhancing healthcare access and appointment management for citizens.

Jirabhron et al. [7] Through the application, patients receive automated reminders of their upcoming appointments, helping them stay informed and reducing the likelihood of forgetting or missing their scheduled visits. The system operates on multiple mobile platforms, making it accessible to a wide range of users.

Dataset

We gather information from the data website names **Kaggle** that already maintain essential records for machine learning.

Links:

- **Fee Prediction:**

We used around 42,000 medical data for implementing the “Fee Prediction” model.

<https://www.kaggle.com/code/sagnik1511/doctor-s-fee-prediction-using-random-forest>

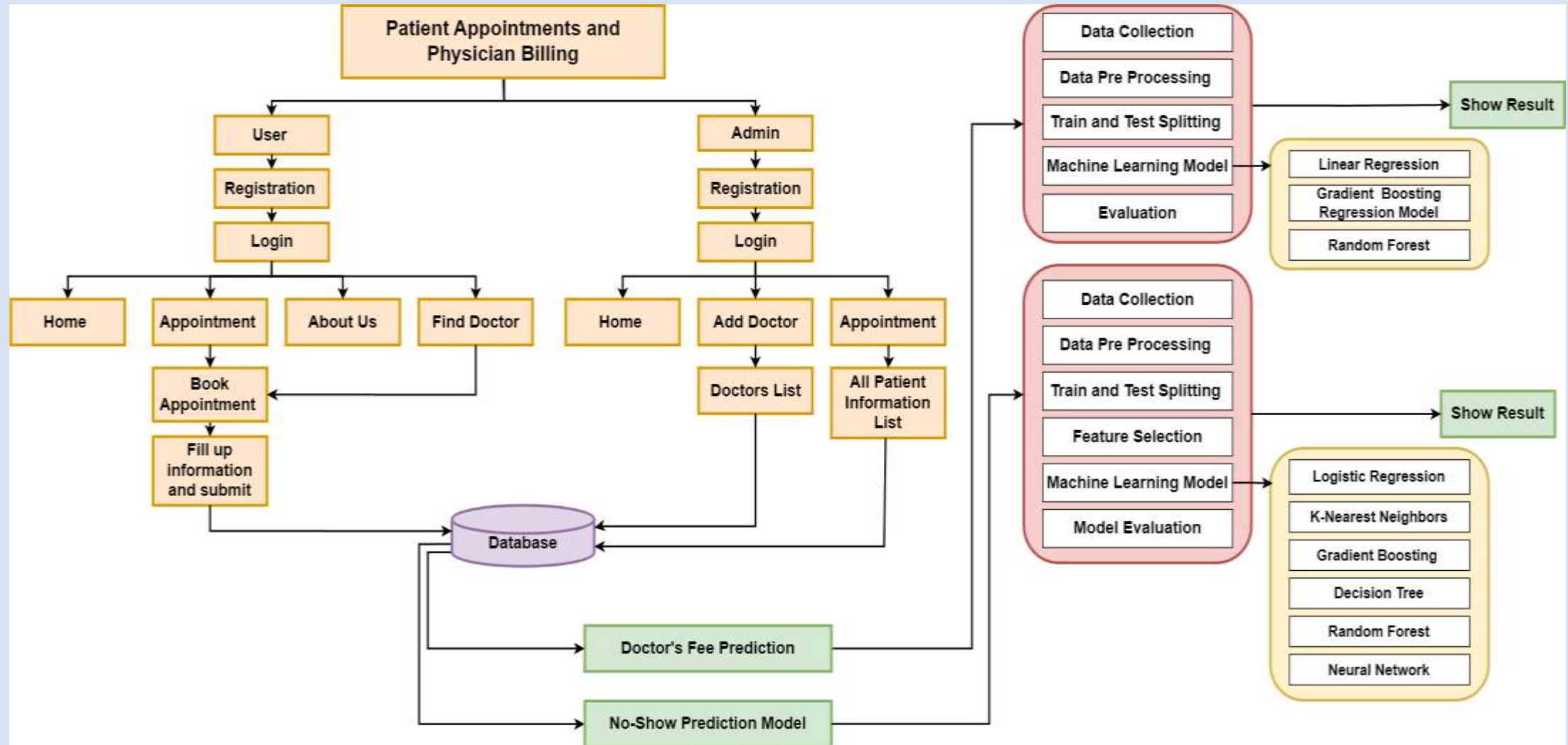
- **No-Show Model:**

We used around 15,50,000 medical data for implementing the “No-Show” model.

<https://www.kaggle.com/datasets/joniarroba/noshowappointments/code?datasetId=792&sortBy=voteCount>



Proposed Methodology



Result Analysis (Doctor's Fee Prediction System)

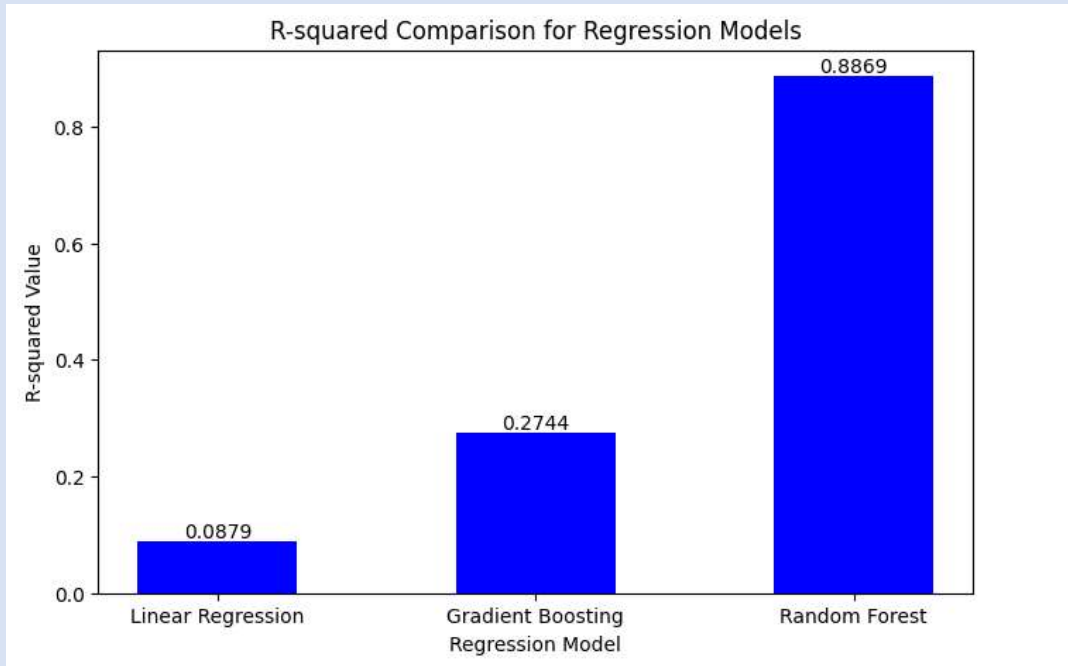


Figure : Comparison of R-squared value

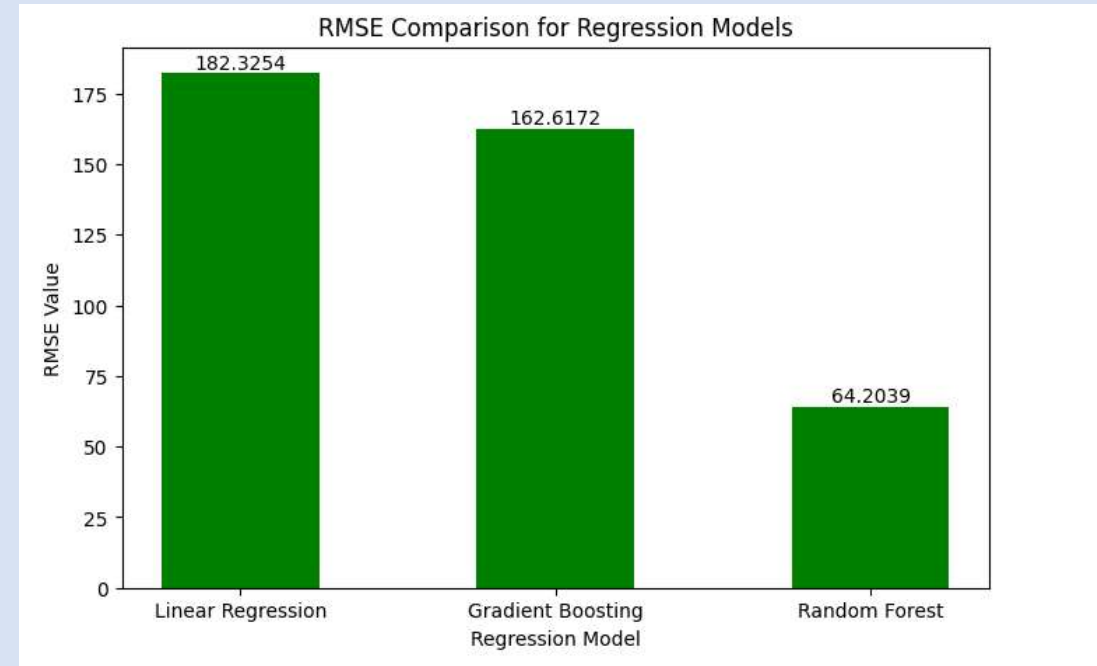


Figure : Comparison of RMSE Value

Result Analysis (No Show Model)

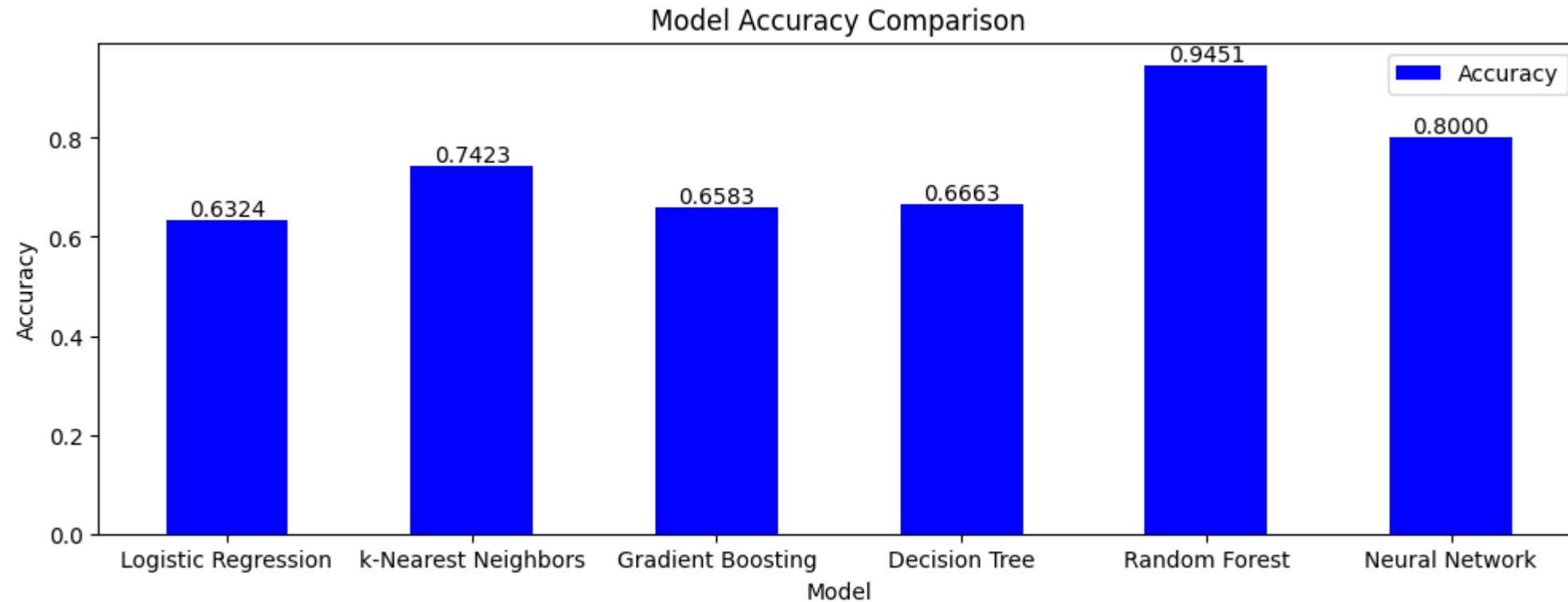


Figure : Comparison of no show model accuracy

Future Work

- Implement automatic retraining of the Machine Learning model method to adapt to dynamic patient behavior.



Conclusion

- Doctor Portal reduces the troubles of people.
- People do not need to waste their one day by going to hospital for booking an appointment.
- Provides an adaptive and intelligent scheduling solution.



Reference

- [3]Yavuz Inal (Department of Information Science and Media Studies, University of Bergen, Bergen, Norway)
- [7]J. Chaiwongsai Also published under: Jira Born Chaibongsai, Jirabhron Chaiwongsai School of Information and Communication Technology,University of Phayao,Phayao, Thailand



**THANK
YOU**